SECTION 2 — TECHNICAL

2.1 Architecture Overview

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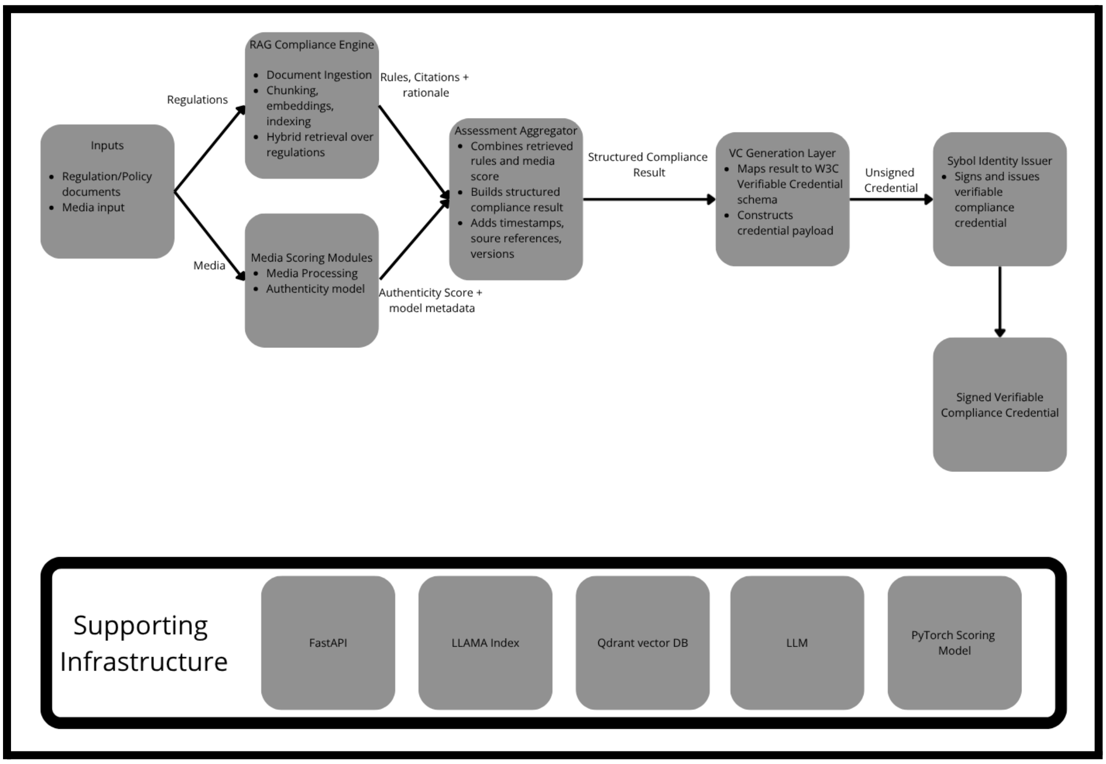
The system is composed of four core components that work together to score media and produce a verifiable compliance credential:

- **RAG Compliance Engine** — ingests EU regulation documents (AI Act, GDPR, DPP, LOPDGDD), chunks and indexes them into a vector database, and retrieves relevant compliance rules and article citations for a given media asset or query
- **Media Scoring Module** — takes an image as input, runs signal extraction across four independent dimensions (metadata integrity, generative artifact detection, visual consistency, provenance), assembles a feature vector, and outputs a calibrated authenticity score between 0 and 1
- **VC Generation Layer** — encodes the authenticity score, per-signal breakdown, compliance status, and regulation citations into a W3C VC Data Model 2.0 credential payload, structured according to Sybol's catalog schema
- **Sybol Identity Integration** — submits the unsigned credential payload to Sybol's businessLogic API, which signs and issues the credential via Sybol's DID infrastructure and svault cryptographic layer, returning a fully signed Verifiable Credential

Supporting Infrastructure: FastAPI | LlamaIndex | Qdrant vector DB | Mistral Large | HuggingFace Models



2.2 Tech Stack

Layer	Tool / Framework	Status
RAG Pipeline	LlamaIndex	Confirmed
LLM	Mistral Large (Mistral AI API)	Confirmed
Embedding Model	sentence-transformers/all-MiniLM-L6-v2	Confirmed
Vector DB	Qdrant	Confirmed
Media Analysis	HuggingFace + OpenCV + ExifRead + imagehash	Confirmed
VC Schema	W3C VC Data Model 2.0	Confirmed
Backend	FastAPI	Confirmed
Deployment	Railway	Confirmed

LlamaIndex

- Orchestrates the full RAG loop end-to-end
- Handles PDF ingestion, chunking, embedding, Qdrant indexing, and query
- Preserves article number and regulation name as chunk metadata
- Native async interface, integrates directly with FastAPI

Mistral Large

- European provider — supports the EU compliance narrative of the paper
- Favourable GDPR data processing terms vs. US providers (OpenAI, Anthropic)
- Competitive with GPT-4o on structured legal text
- Accessed via Mistral AI API; no self-hosting required

sentence-transformers/all-MiniLM-L6-v2

- Runs fully locally — no external API call, no data transfer
- Strongest GDPR posture for the embedding layer
- 384-dimensional dense vectors
- Native LlamaIndex integration out of the box

Qdrant

- Supports hybrid search (dense + sparse) and metadata filtering
- Filter retrieval by regulation type or article number at query time
- Same config for local dev and Railway production — no changes needed
- Stores media scoring audit trail as payload alongside embeddings

HuggingFace — dima806/deepfake_vs_real_image_detection

- Pre-trained CNN for generative artifact detection (signal a)
- Large diverse training dataset of real and synthetic images
- Immediately usable via transformers pipeline — no custom training needed
- Documentable as a validated baseline model in the research paper

OpenCV

- Lighting uniformity, shadow direction, edge blending artifact checks (signal v)
- Facial geometry and landmark detection explicitly excluded
- Architectural decision to avoid biometric data classification (GDPR Art. 4(14), AI Act Art. 5)

ExifRead

- Extracts EXIF metadata before any image transformation (signal m)
- Flags missing camera fields, editing software tags, timestamp anomalies

imagehash (pHash)

- Perceptual hash comparison against known authentic source dataset (signal p)
- No third-party reverse lookup — avoids GDPR joint controllership (Arts. 28, 44)

FastAPI

- Async Python, native integration with the full ML stack
- Automatic OpenAPI schema generation
- Exposes three endpoints: image upload, RAG query, VC issuance trigger

Railway

- Single Docker container deployment
- Secrets managed via Railway secret store
- Qdrant runs as a separate Railway service within the same project

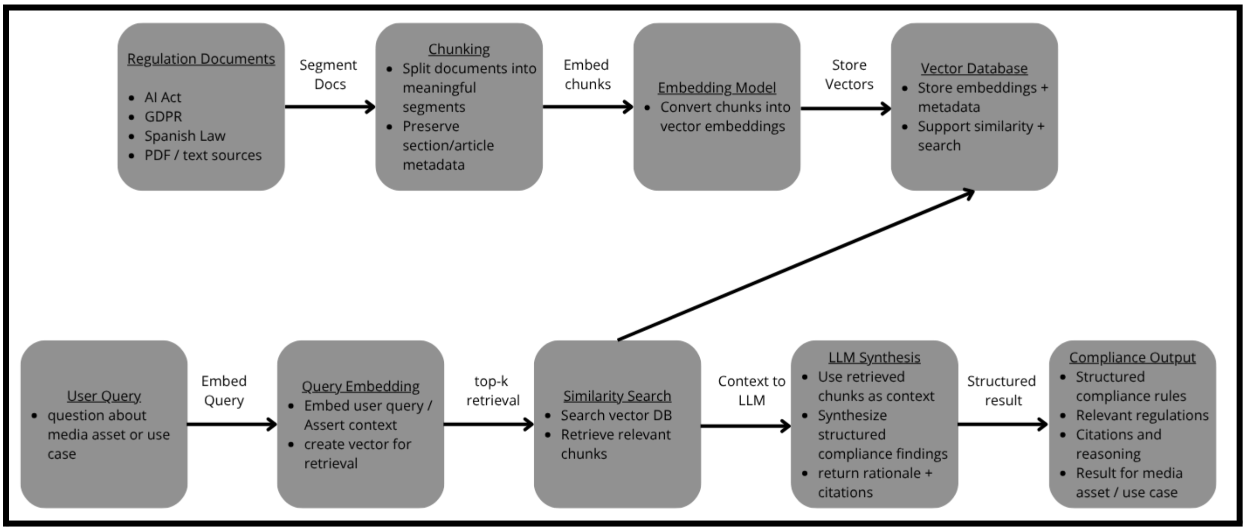
2.3 RAG Pipeline Design

Input Regulation PDFs: EU AI Act, GDPR, ESPR/DPP (EU 2024/1781), LOPDGDD, Ley 13/2022

Processing Steps

1. Parse PDFs with SimpleDirectoryReader (PyMuPDF); attach regulation name, article number, section as chunk metadata
2. Chunk at article level (400–600 tokens) with overlap using SentenceSplitter
3. Embed locally with all-MiniLM-L6-v2; store vectors + metadata in Qdrant
4. On query: embed query → similarity search (top-k = 5) → assemble context → Mistral Large synthesis
5. Output: structured compliance result with regulation rules, article citations, and a regulationRefs array ready for the VC

Output Structured compliance result: regulation rules, article citations (name + number + URL), rationale, regulationRefs array



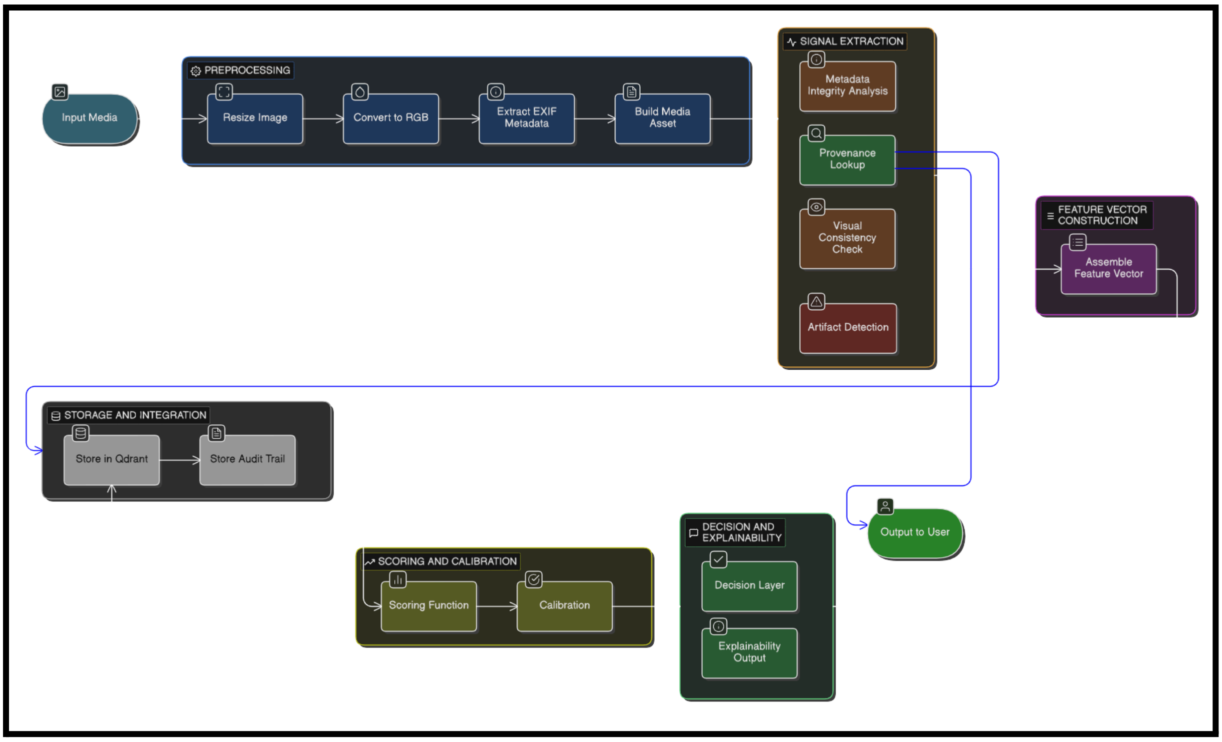
2.4 Media Scoring Module

Input Images only — JPEG, PNG, WebP. Video deferred.

Processing Steps

- Normalisation:
    1. Resize to 224×224 for model input; preserve original for hash
    2. Convert to RGB
    3. Extract EXIF with ExifRead before any transformation
- Signal extraction:
    1. **m — Metadata integrity:** missing EXIF fields, editing software tags (Photoshop, GIMP, SD), timestamp anomalies
    2. **a — Generative artifact:** HuggingFace CNN detector + FFT frequency analysis + noise residual analysis
    3. **v — Visual consistency:** lighting uniformity, shadow direction, edge blending (OpenCV). Facial landmarks explicitly excluded — architectural decision to avoid biometric data classification (GDPR Art. 4(14), AI Act Art. 5)
    4. **p — Provenance:** pHash vs. known authentic dataset. No third-party reverse lookup (GDPR Arts. 28, 44)
- Feature vector: X = [m, a, v, p]
- Scoring: S = wm·m + wa·a + wv·v + wp·p, calibrated via Platt scaling
- Interpretation: 0.0–0.3 → non-compliant | 0.3–0.7 → review | 0.7–1.0 → compliant
- Output stored in Qdrant as audit trail; URL becomes evidenceUrl in the VC

Output Float 0.0 → 1.0 + signal breakdown [m, a, v, p] + complianceStatus



2.5 Verifiable Credential Schema

Field	Description
id	UUID URN for the credential
type	["VerifiableCredential", "MediaComplianceCredential"]
issuer	Sybol DID — confirm with Iñigo
validFrom	ISO 8601 issuance timestamp
credentialSchema	Sybol catalog schema reference
credentialStatus	StatusList2021 revocation entry
credentialSubject.id	urn:media:{sha256}
credentialSubject.mediaHash	SHA-256 of raw file before resizing
credentialSubject.authenticityScore	Float 0.0 → 1.0
credentialSubject.scoreBreakdown	Per-signal scores: m, a, v, p
credentialSubject.complianceStatus	compliant / non-compliant / review

credentialSubject.regulationRefs	Structured array: regulation, article, URL
credentialSubject.modelVersion	Model version for traceability
credentialSubject.analysisTimestamp	When analysis was run
credentialSubject.evidenceUrl	Link to Qdrant audit trail record

To confirm with Iñigo:

- Sybol issuer DID value
- Catalog registration flow for MEDIA_COMPLIANCE_CREDENTIAL
- Signing flow: do we call businessLogic /credentials/issue directly, or a separate signing endpoint?